

Practical Demo: Vasicek Rate Scenarios and Long-Run Behavior

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Purpose and Assumptions

The Vasicek model is

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t.$$

This demo estimates its parameters from a dated monthly U.S. effective federal funds-rate history, simulates scenarios, checks long-run moments, and uses the fitted distribution for a one-year risk decision. The embedded observations are monthly averages for January 2023

through June 2025, recorded in percent and converted to decimals. Source: [Federal Reserve Bank of St. Louis, Effective Federal Funds Rate \(FEDFUNDS\)](#). Parameters are constant; estimation error, regime shifts, jumps, and term premia are ignored. This short sample is pedagogical and should not be treated as a robust trading calibration.

1 Step 1: Record the observed rate history

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

dates = pd.date_range("2023-01-31", periods=30, freq="ME")
rates = np.array([
    4.33, 4.57, 4.65, 4.83, 5.06, 5.08,
    5.12, 5.33, 5.33, 5.33, 5.33, 5.33,
    5.33, 5.33, 5.33, 5.33, 5.33, 5.33,
    5.33, 5.33, 5.13, 4.83, 4.64, 4.48,
    4.33, 4.33, 4.33, 4.33, 4.33, 4.33,
]) / 100
history = pd.Series(rates, index=dates, name="Effective federal funds rate")
history.to_frame().tail()
```

	Effective federal funds rate
2025-02-28	0.0433
2025-03-31	0.0433
2025-04-30	0.0433
2025-05-31	0.0433
2025-06-30	0.0433

The series shows tightening, a long plateau, and initial easing. It is too short and policy-driven for a robust structural calibration, but it lets us show the entire estimation workflow without hiding the model's limitations.

2 Step 2: Estimate Vasicek parameters

The exact monthly transition can be written as an AR(1):

$$r_{t+\Delta} = \alpha + \phi r_t + \varepsilon_{t+\Delta}, \quad \phi = e^{-\kappa\Delta}, \quad \alpha = \theta(1 - \phi).$$

After OLS estimation,

$$\hat{\kappa} = -\frac{\log \hat{\phi}}{\Delta}, \quad \hat{\theta} = \frac{\hat{\alpha}}{1 - \hat{\phi}}, \quad \hat{\sigma} = s_\varepsilon \sqrt{\frac{2\hat{\kappa}}{1 - \hat{\phi}^2}}.$$

Here $\Delta = 1/12$ year, κ is in 1/year, and σ is in decimal rate units per $\sqrt{\text{year}}$.

```
dt = 1 / 12
x = history.iloc[:-1].to_numpy()
y = history.iloc[1:].to_numpy()
design = np.column_stack([np.ones_like(x), x])
alpha_hat, phi_hat = np.linalg.lstsq(design, y, rcond=None)[0]
residuals = y - (alpha_hat + phi_hat * x)
residual_std = residuals.std(ddof=2)

if not 0 < phi_hat < 1:
    raise ValueError("This sample does not imply a stationary Vasicek fit.")

kappa = -np.log(phi_hat) / dt
theta = alpha_hat / (1 - phi_hat)
sigma = residual_std * np.sqrt(2 * kappa / (1 - phi_hat**2))
r0 = float(history.iloc[-1])

half_life = np.log(2) / kappa
long_run_std = sigma / np.sqrt(2 * kappa)

pd.Series({
    "Current rate": r0,
    "Estimated monthly persistence phi": phi_hat,
    "Mean-reversion speed kappa (1/year)": kappa,
    "Long-run mean theta": theta,
    "Volatility sigma (rate/sqrt(year))": sigma,
    "Mean-reversion half-life (years)": half_life,
    "Long-run standard deviation": long_run_std,
}).round(5)
```

Current rate	0.0433
Estimated monthly persistence phi	0.9578

Mean-reversion speed kappa (1/year)	0.5173
Long-run mean theta	0.0495
Volatility sigma (rate/sqrt(year))	0.0043
Mean-reversion half-life (years)	1.3399
Long-run standard deviation	0.0042
dtype: float64	

- κ controls how quickly shocks decay; the half-life is $\ln(2)/\kappa$.
- θ is the long-run mean under the chosen probability measure.
- σ controls the size of random innovations and the long-run dispersion.

3 Simulate Monthly Paths

The exact Vasicek transition is normal:

$$r_{t+\Delta} = \theta + (r_t - \theta)e^{-\kappa\Delta} + \sigma\sqrt{\frac{1 - e^{-2\kappa\Delta}}{2\kappa}}Z.$$

```
rng = np.random.default_rng(411)

def simulate_vasicek(initial_rate, years, paths, rng):
    steps = int(round(years / dt))
    output = np.empty((steps + 1, paths))
    output[0] = initial_rate
    persistence = np.exp(-kappa * dt)
    innovation_std = sigma * np.sqrt((1 - np.exp(-2 * kappa * dt)) / (2 * kappa))
    for t in range(steps):
        output[t + 1] = (
            theta + (output[t] - theta) * persistence
            + innovation_std * rng.standard_normal(paths)
        )
    return output

paths = simulate_vasicek(r0, years=5, paths=5_000, rng=rng)
time = np.arange(paths.shape[0]) * dt
```

```
fig, ax = plt.subplots(figsize=(7.5, 4.1))
for i in range(30):
    ax.plot(time, 100 * paths[:, i], alpha=0.28, linewidth=0.8)
ax.plot(time, 100 * paths.mean(axis=1), color="black", linewidth=2.2, label="Simulation mean")
```

```

ax.axhline(100 * theta, color="#F58518", linestyle="--", label="Long-run mean")
ax.set(xlabel="Years", ylabel="Short rate (%)", title="Illustrative Vasicek scenarios")
ax.grid(alpha=0.3)
ax.legend()
plt.tight_layout()
plt.show()

```

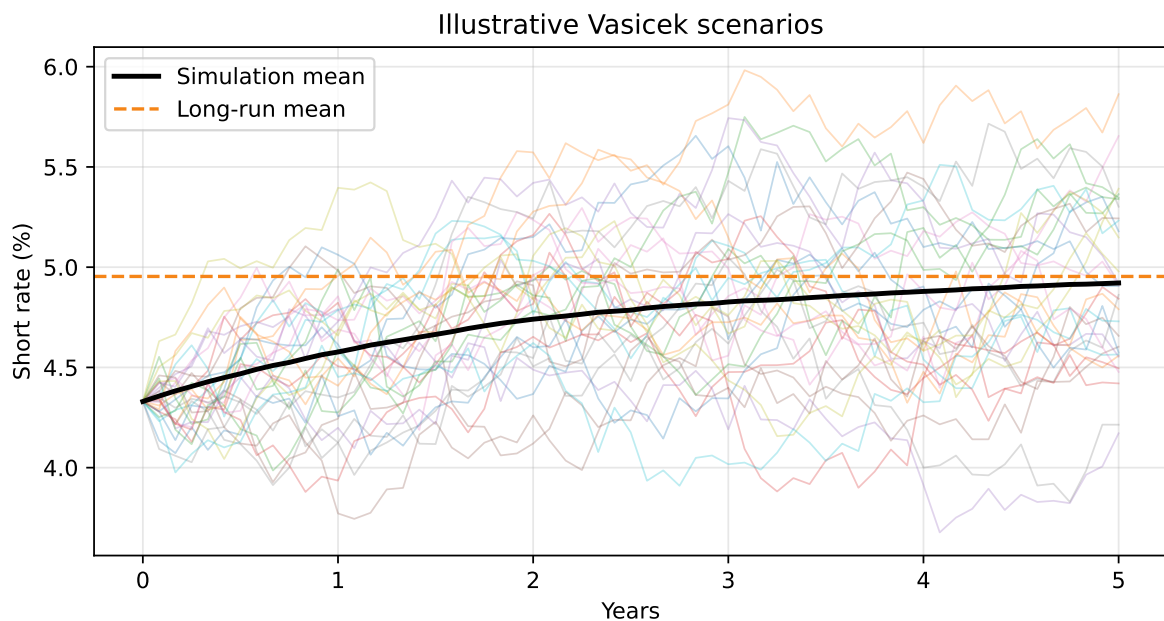


Figure 1: Vasicek paths fluctuate randomly but revert toward the long-run mean

4 Distribution and Negative Rates

```

horizon_rate = paths[-1]
summary = pd.Series({
    "Five-year mean": horizon_rate.mean(),
    "Five-year standard deviation": horizon_rate.std(ddof=1),
    "5th percentile": np.quantile(horizon_rate, 0.05),
    "Median": np.median(horizon_rate),
    "95th percentile": np.quantile(horizon_rate, 0.95),
    "Probability of negative rate": np.mean(horizon_rate < 0),
})
summary.round(5)

```

Five-year mean	0.0492
Five-year standard deviation	0.0042
5th percentile	0.0421
Median	0.0492
95th percentile	0.0560
Probability of negative rate	0.0000
dtype: float64	

Vasicek rates are normally distributed, so negative rates are possible. That can be useful in markets where negative policy rates are plausible, but it may be undesirable for other applications. A positivity-preserving model such as CIR changes this feature, at the cost of a different distribution and dynamics.

5 Use the fitted distribution in a funding decision

Suppose a treasury team wants a one-year planning range for a floating-rate funding cost tied to the overnight rate. Simulate 50,000 one-year endpoints and report the central 90% range and the probability that the rate exceeds 5%. This applies the fitted **physical-measure** model; it is not a risk-neutral bond price.

```
one_year_rates = simulate_vasicek(
    r0, years=1.0, paths=50_000, rng=np.random.default_rng(414)
)[-1]
funding_summary = pd.Series({
    "5th-percentile funding rate (%)": 100 * np.quantile(one_year_rates, 0.05),
    "Median funding rate (%)": 100 * np.median(one_year_rates),
    "95th-percentile funding rate (%)": 100 * np.quantile(one_year_rates, 0.95),
    "Probability funding rate exceeds 5%": np.mean(one_year_rates > 0.05),
})
funding_summary.round(3)
```

5th-percentile funding rate (%)	4.023
Median funding rate (%)	4.581
95th-percentile funding rate (%)	5.137
Probability funding rate exceeds 5%	0.107
dtype: float64	

The 95th percentile can serve as a planning stress, while the exceedance probability helps size an interest-expense buffer. Parameter uncertainty and policy regime changes should be added as separate stresses.

6 Model Scenario Versus a Historical Shock

Use the largest one-month decline in the compact history as a simple historical shock, and compare it with the model's one-month distribution.

```
monthly_changes = history.diff().dropna()
historical_down_shock = monthly_changes.min()
model_one_month = simulate_vasicek(r0, years=dt, paths=50_000,
                                   rng=np.random.default_rng(412))[-1] - r0

pd.Series({
    "Largest historical monthly decline (bp)": historical_down_shock * 10_000,
    "Model mean one-month change (bp)": model_one_month.mean() * 10_000,
    "Model 5th-percentile change (bp)": np.quantile(model_one_month, 0.05) * 10_000,
    "Model 1st-percentile change (bp)": np.quantile(model_one_month, 0.01) * 10_000,
}).round(2)
```

Largest historical monthly decline (bp)	-30.00
Model mean one-month change (bp)	2.67
Model 5th-percentile change (bp)	-17.46
Model 1st-percentile change (bp)	-25.37

dtype: float64

The historical scenario preserves an observed move but supplies only one outcome. The Vasicek simulation supplies a distribution and mean reversion, but depends on model specification and parameters. Risk management should compare both rather than treating either as truth.

7 Validate Long-Run Moments and Units

The stationary Vasicek distribution has

$$E[r] = \theta, \quad \text{Var}(r) = \frac{\sigma^2}{2\kappa}.$$

```
long_paths = simulate_vasicek(r0, years=20, paths=30_000,
                              rng=np.random.default_rng(413))[-1]
simulated_mean = long_paths.mean()
simulated_std = long_paths.std(ddof=1)
```

```

assert kappa > 0 and sigma > 0
assert abs(simulated_mean - theta) < 0.0005
assert abs(simulated_std - long_run_std) < 0.0005
assert abs((0.0001 * 10_000) - 1.0) < 1e-12

pd.Series({
    "Theoretical long-run mean": round(theta, 5),
    "Simulated 20-year mean": round(simulated_mean, 5),
    "Theoretical long-run standard deviation": round(long_run_std, 5),
    "Simulated 20-year standard deviation": round(simulated_std, 5),
    "Decimal 0.0001 equals one basis point": True,
})

```

```

Theoretical long-run mean          0.0495
Simulated 20-year mean             0.0496
Theoretical long-run standard deviation  0.0042
Simulated 20-year standard deviation    0.0043
Decimal 0.0001 equals one basis point    True
dtype: object

```

8 Financial Interpretation

Mean reversion makes the expected rate move toward θ , while volatility keeps future rates dispersed. κ determines how quickly today's rate loses influence; σ determines scenario width. The model is a disciplined scenario generator, not a forecast guarantee. Calibration window, probability measure, structural breaks, and the normal model's negative-rate tail all affect how its output should be used.